# CO3 & CO4

# PROGRAM – 1

**Design a Flutter application that displays a list of products with product images, names, prices, ratings, and an "Add to Cart" button.**

## 1. Aim

To design and develop a Flutter application that displays a list of products with **product images, product names, prices, ratings, and an "Add to Cart" button**.

## 2. Objective

- To understand the basic structure of a Flutter application.
- To create a user interface using Flutter widgets.
- To display multiple products using ListView.builder.
- To display product images using Image.network().
- To arrange product details using Row and Column.
- To implement an **Add to Cart** button.
- To display a SnackBar when a product is added to the cart.

## 3. Software Requirements

- Flutter SDK
- Dart
- Android Studio / Visual Studio Code
- Android Emulator or physical Android device
- Internet connection for loading network images

## 4. Widgets Used

| Widget | Purpose |
|---|---|
| MaterialApp | Provides the basic Material Design structure |
| Scaffold | Provides the basic screen structure |
| AppBar | Displays the application title |
| ListView.builder | Displays the list of products |
| Card | Creates a card-like container for each product |
| Row | Arranges widgets horizontally |
| Column | Arranges widgets vertically |
| Image.network() | Displays images from URLs |
| Text | Displays product information |
| ElevatedButton | Creates the Add to Cart button |
| SnackBar | Displays a temporary message |
| Expanded | Uses the remaining available horizontal space |

## 5. Procedure

1. Start the Flutter application.
2. Import the material.dart package.
3. Create the main() function and call runApp().
4. Create the MyApp widget.
5. Configure MaterialApp with a title, theme, and ProductPage.
6. Create a list containing product details such as:
    - Product name
    - Price

- Rating
- Image URL

7. Create a Scaffold containing an AppBar.
8. Use ListView.builder() to display all products.
9. For each product, create a Card.
10. Display the product image using Image.network().
11. Display the product name, price, and rating using Text widgets.
12. Add an ElevatedButton with the text **"Add to Cart"**.
13. When the button is pressed, display a SnackBar showing that the product has been added to the cart.
14. Run the application and verify the output.
15. Stop.

**6. Program**

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
```

```
      debugShowCheckedModeBanner: false,
      title: 'Product List',
      theme: ThemeData(
        primarySwatch: Colors.blue,
      ),
      home: const ProductPage(),
    );
  }
}

class ProductPage extends StatelessWidget {
  const ProductPage({super.key});

  final List<Map<String, dynamic>> products = const [
    {
      'name': 'Smartphone',
      'price': 19999,
      'rating': 4.5,
      'image':
          'https://images.unsplash.com/photo-1511707171634-
5f897ff02aa9?w=500',
    },
    {
      'name': 'Laptop',
      'price': 59999,
      'rating': 4.7,
```

```
    'image':
        'https://images.unsplash.com/photo-1496181133206-
80ce9b88a853?w=500',
    },
    {
     'name': 'Headphones',
     'price': 2999,
     'rating': 4.3,
     'image':
        'https://images.unsplash.com/photo-1505740420928-
5e560c06d30e?w=500',
    },
    {
     'name': 'Smart Watch',
     'price': 4999,
     'rating': 4.4,
     'image':
        'https://images.unsplash.com/photo-1523275335684-
37898b6baf30?w=500',
    },
  ];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Product List'),
```

```
  centerTitle: true,
),
body: ListView.builder(
  padding: const EdgeInsets.all(10),
  itemCount: products.length,
  itemBuilder: (context, index) {
    final product = products[index];

    return Card(
      margin: const EdgeInsets.only(bottom: 12),
      elevation: 4,
      child: Padding(
        padding: const EdgeInsets.all(10),
        child: Row(
          children: [
            ClipRRect(
              borderRadius: BorderRadius.circular(10),
              child: Image.network(
                product['image'],
                width: 100,
                height: 100,
                fit: BoxFit.cover,
              ),
            ),

            const SizedBox(width: 15),
```

```
Expanded(
  child: Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [
      Text(
        product['name'],
        style: const TextStyle(
          fontSize: 20,
          fontWeight: FontWeight.bold,
        ),
      ),

      const SizedBox(height: 8),

      Text(
        '₹${product['price']}',
        style: const TextStyle(
          fontSize: 18,
          color: Colors.green,
          fontWeight: FontWeight.bold,
        ),
      ),

      const SizedBox(height: 5),
```

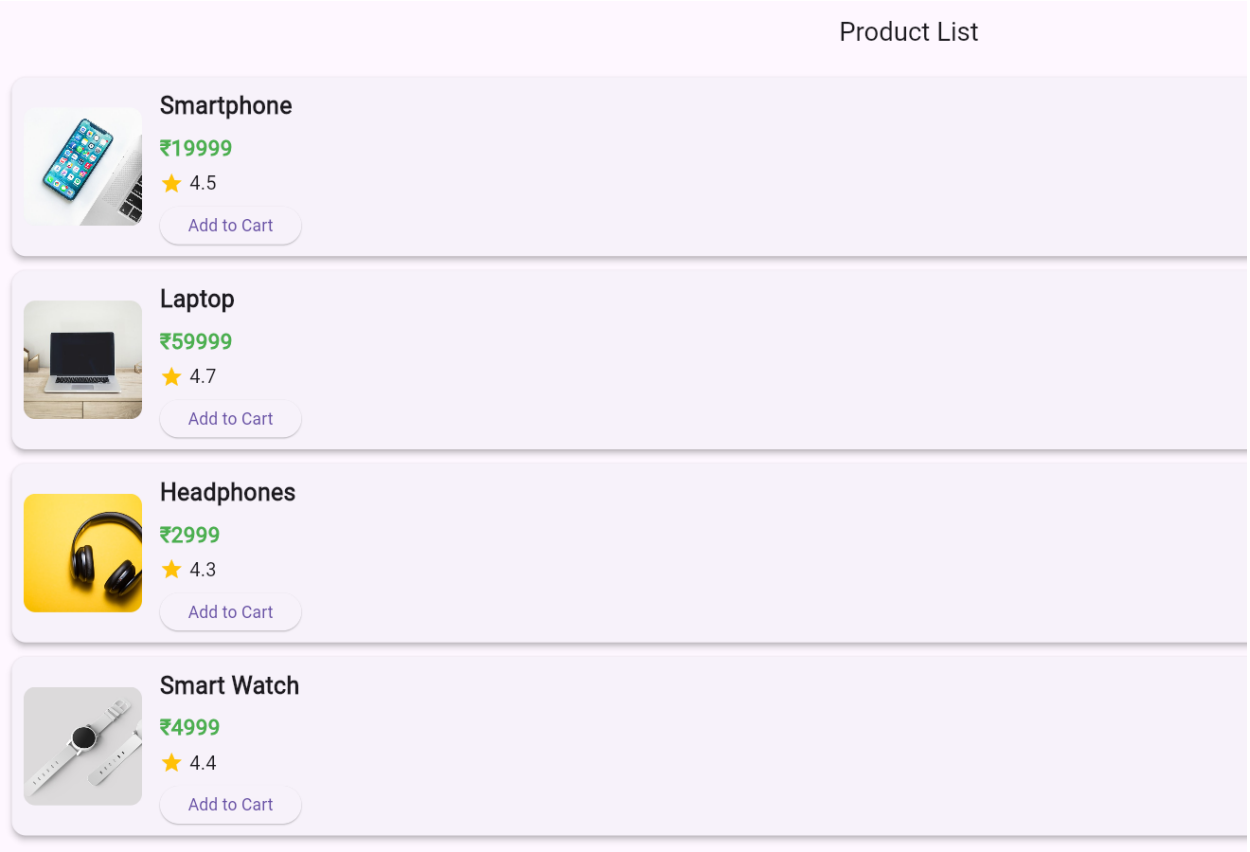
```
Row(
  children: [
    const Icon(
      Icons.star,
      color: Colors.amber,
      size: 20,
    ),
    const SizedBox(width: 5),
    Text(
      '${product['rating']}',
      style: const TextStyle(fontSize: 16),
    ),
  ],
),

const SizedBox(height: 8),

ElevatedButton(
  onPressed: () {
    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(
        content: Text(
          '${product['name']} added to cart',
        ),
      ),
    );
```

```
                    },
                    child: const Text('Add to Cart'),
                  ),
                ],
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

## 7.Output




## 8. Result

Thus, a Flutter application was successfully developed to display a list of products with **images, names, prices, ratings, and an Add to Cart button**. The application also successfully displays a confirmation message using a SnackBar when a product is added to the cart.

## PROGRAM – 2

**Design a Flutter application that displays a circular profile image with the user's name and designation. When the user taps the profile image, it should smoothly enlarge, change its border color, and display additional profile details (Email and Phone Number). Tapping the image again should restore the original size and hide the additional details. Include an animated "Follow" button that changes to "Following" when pressed.**

### Flutter Profile Application

### 1. Aim

To design and develop a Flutter profile application that displays a profile picture, name, designation, contact details, and an interactive Follow button with animations.

### 2. Objective

- To understand StatefulWidget and state management.
- To use setState() for updating the UI.
- To display a local image using Image.asset().
- To implement animations using AnimatedContainer and AnimatedSwitcher.
- To handle user interaction using GestureDetector.

### 3. Software Requirements

- Flutter SDK
- Dart
- Android Studio / Visual Studio Code
- Android Emulator or Android device

### 4. Widgets Used

- MaterialApp – Application structure
- Scaffold – Basic screen layout

- AppBar – Displays the title
- StatefulWidget – Handles dynamic UI
- GestureDetector – Detects tap gestures
- AnimatedContainer – Animates profile image size and border
- AnimatedSwitcher – Animates changing widgets
- Image.asset() – Displays the local profile image
- ClipOval – Makes the image circular
- ElevatedButton – Creates the Follow button
- Text – Displays profile information

**5. Procedure**

1. Create a new Flutter project.
2. Add the profile image as assets/profile.jpg.
3. Declare the asset in pubspec.yaml.
4. Import the Flutter Material package.
5. Create a StatefulWidget named ProfilePage.
6. Create Boolean variables isExpanded and isFollowing.
7. Use GestureDetector to detect taps on the profile image.
8. Use AnimatedContainer to enlarge the profile image when tapped.
9. Use AnimatedSwitcher to display the email and phone number.
10. Use an ElevatedButton to toggle between **Follow** and **Following**.
11. Run the application and verify the output.

**6. Program**

```
import 'package:flutter/material.dart';


void main() {
  runApp(const MyApp());
```

```
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Profile App',
      theme: ThemeData(
        primarySwatch: Colors.blue,
      ),
      home: const ProfilePage(),
    );
  }
}

class ProfilePage extends StatefulWidget {
  const ProfilePage({super.key});

  @override
  State<ProfilePage> createState() => _ProfilePageState();
}

class _ProfilePageState extends State<ProfilePage> {
```

```
bool isExpanded = false;
bool isFollowing = false;

void toggleProfile() {
  setState(() {
    isExpanded = !isExpanded;
  });
}

void toggleFollow() {
  setState(() {
    isFollowing = !isFollowing;
  });
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Profile'),
      centerTitle: true,
    ),
    body: Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
```

```
GestureDetector(
  onTap: toggleProfile,
  child: AnimatedContainer(
    duration: const Duration(milliseconds: 500),
    width: isExpanded ? 180 : 120,
    height: isExpanded ? 180 : 120,
    padding: const EdgeInsets.all(5),
    decoration: BoxDecoration(
      shape: BoxShape.circle,
      border: Border.all(
        color: isExpanded ? Colors.blue : Colors.grey,
        width: 5,
      ),
    ),
    child: ClipOval(
      child: Image.asset(
        'assets/profile.jpg',
        fit: BoxFit.cover,
      ),
    ),
  ),
),

const SizedBox(height: 20),

const Text(
```

```
  'R PRANAV',
  style: TextStyle(
    fontSize: 26,
    fontWeight: FontWeight.bold,
  ),
),

const SizedBox(height: 5),

const Text(
  'Software Developer',
  style: TextStyle(
    fontSize: 18,
    color: Colors.grey,
  ),
),

const SizedBox(height: 20),

AnimatedSwitcher(
  duration: const Duration(milliseconds: 400),
  child: isExpanded
      ? Column(
          key: const ValueKey('details'),
          children: const [
            Text(
```

```
            'Email: pranav@gmail.com',
            style: TextStyle(fontSize: 16),
          ),
          SizedBox(height: 8),
          Text(
            'Phone: 9876543210',
            style: TextStyle(fontSize: 16),
          ),
        ],
      )
    : const SizedBox(
        key: ValueKey('empty'),
      ),
),

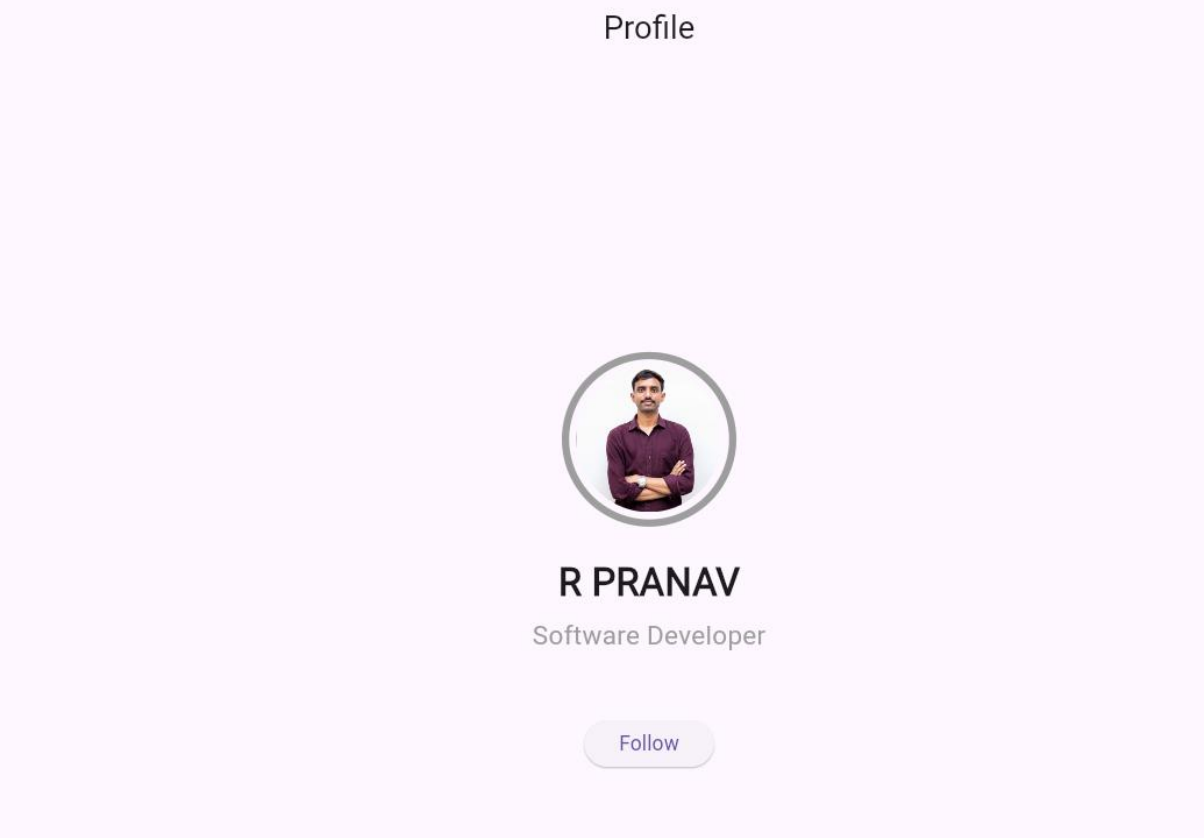
const SizedBox(height: 25),

AnimatedSwitcher(
  duration: const Duration(milliseconds: 300),
  child: ElevatedButton(
    key: ValueKey(isFollowing),
    onPressed: toggleFollow,
    child: Text(
      isFollowing ? 'Following' : 'Follow',
    ),
  ),
```

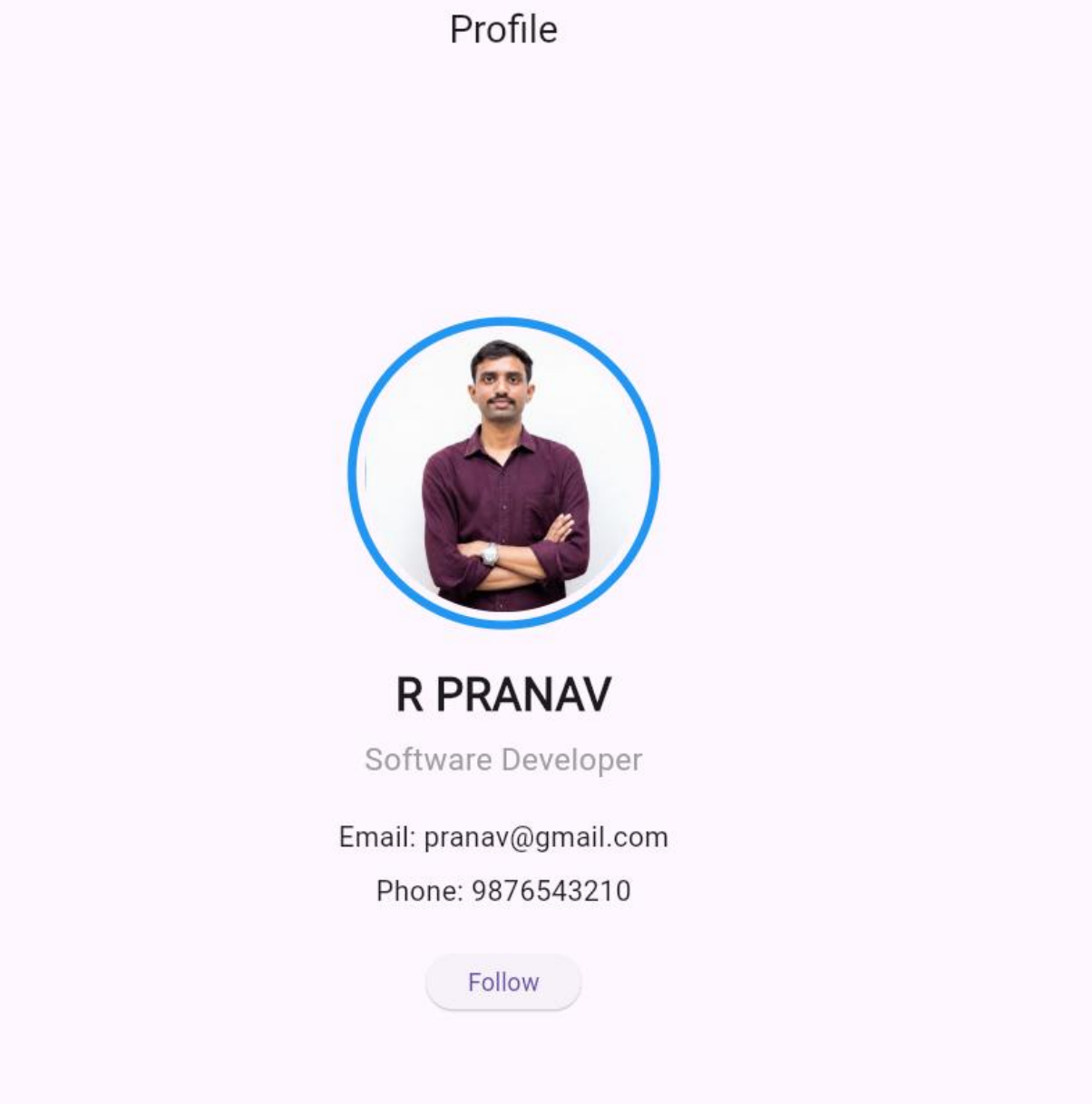
```
            ),
          ],
        ),
      ),
    );
  }
}
```

**7. <u>Output</u>**

**Befor clicking:-**



**After Clicking :-**




## 8. Result

Thus, the Flutter profile application was successfully designed and implemented with interactive elements and animations using StatefulWidget, setState(), GestureDetector, AnimatedContainer, and AnimatedSwitcher.

# PROGRAM – 3

**Create a Flutter application to design a Student Registration Form using the Form widget with validation.**

## 1. Aim

To design and develop a Flutter application for student registration with form fields for name, roll number, email, and phone number along with input validation.

## 2. Objective

- To create a student registration form using Flutter.
- To understand the use of Form and TextFormField.
- To implement form validation using validator.
- To use TextEditingController to manage input fields.
- To display a success message using SnackBar.

## 3. Software Requirements

- Flutter SDK
- Dart
- Android Studio / Visual Studio Code
- Android Emulator or Android device

## 4. Widgets Used

- MaterialApp – Provides the application structure.
- Scaffold – Provides the basic screen layout.
- AppBar – Displays the application title.
- SingleChildScrollView – Allows the form to scroll.
- Form – Groups and validates form fields.
- TextFormField – Accepts student information.
- InputDecoration – Provides labels, icons, and borders.

- ElevatedButton – Creates the Register button.
- SnackBar – Displays registration success message.
- TextEditingController – Manages text input.

**5. Procedure**

1. Create a new Flutter project.
2. Import the Flutter Material package.
3. Create the MyApp widget and configure the application.
4. Create RegistrationPage as a StatefulWidget.
5. Create a GlobalKey<FormState> for form validation.
6. Create TextEditingController objects for name, roll number, email, and phone number.
7. Create a registration form using Form and TextFormField.
8. Add validation for each input field.
9. Add a Register button.
10. When the Register button is pressed, validate the form using validate().
11. If all fields are valid, display **"Registration Successful!"** using a SnackBar.
12. Run the application and verify the output.

**6. Program**

```
import 'package:flutter/material.dart';


void main() {
  runApp(const MyApp());
}


class MyApp extends StatelessWidget {
  const MyApp({super.key});
```

```
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Student Registration',
      theme: ThemeData(
        primarySwatch: Colors.blue,
      ),
      home: const RegistrationPage(),
    );
  }
}

class RegistrationPage extends StatefulWidget {
  const RegistrationPage({super.key});

  @override
  State<RegistrationPage> createState() =>
      _RegistrationPageState();
}

class _RegistrationPageState extends State<RegistrationPage> {

  final _formKey = GlobalKey<FormState>();
```

```
final nameController = TextEditingController();
final emailController = TextEditingController();
final phoneController = TextEditingController();
final rollNoController = TextEditingController();

void registerStudent() {
  if (_formKey.currentState!.validate()) {
    ScaffoldMessenger.of(context).showSnackBar(
      const SnackBar(
        content: Text('Registration Successful!'),
      ),
    );
  }
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Student Registration'),
      centerTitle: true,
    ),

    body: SingleChildScrollView(
      padding: const EdgeInsets.all(20),
```

```
child: Form(
  key: _formKey,

  child: Column(
    crossAxisAlignment: CrossAxisAlignment.stretch,
    children: [

      const Icon(
        Icons.school,
        size: 80,
        color: Colors.blue,
      ),

      const SizedBox(height: 15),

      const Text(
        'Student Registration Form',
        textAlign: TextAlign.center,
        style: TextStyle(
          fontSize: 24,
          fontWeight: FontWeight.bold,
        ),
      ),

      const SizedBox(height: 25),
```

```
TextFormField(
  controller: nameController,
  decoration: const InputDecoration(
    labelText: 'Name',
    prefixIcon: Icon(Icons.person),
    border: OutlineInputBorder(),
  ),
  validator: (value) {
    if (value == null || value.trim().isEmpty) {
      return 'Please enter your name';
    }
    return null;
  },
),

const SizedBox(height: 15),

TextFormField(
  controller: rollNoController,
  decoration: const InputDecoration(
    labelText: 'Roll Number',
    prefixIcon: Icon(Icons.badge),
    border: OutlineInputBorder(),
  ),
  validator: (value) {
    if (value == null || value.trim().isEmpty) {
```

```
      return 'Please enter roll number';
    }
    return null;
  },
),

const SizedBox(height: 15),

TextFormField(
  controller: emailController,
  keyboardType: TextInputType.emailAddress,
  decoration: const InputDecoration(
    labelText: 'Email',
    prefixIcon: Icon(Icons.email),
    border: OutlineInputBorder(),
  ),
  validator: (value) {
    if (value == null || value.trim().isEmpty) {
      return 'Please enter email';
    }

    if (!value.contains('@')) {
      return 'Enter a valid email';
    }

    return null;
```

```
  },
),

const SizedBox(height: 15),

TextFormField(
  controller: phoneController,
  keyboardType: TextInputType.phone,
  decoration: const InputDecoration(
    labelText: 'Phone Number',
    prefixIcon: Icon(Icons.phone),
    border: OutlineInputBorder(),
  ),
  validator: (value) {
    if (value == null || value.trim().isEmpty) {
      return 'Please enter phone number';
    }

    if (value.length != 10) {
      return 'Phone number must contain 10 digits';
    }

    return null;
  },
),
```

```
            const SizedBox(height: 25),

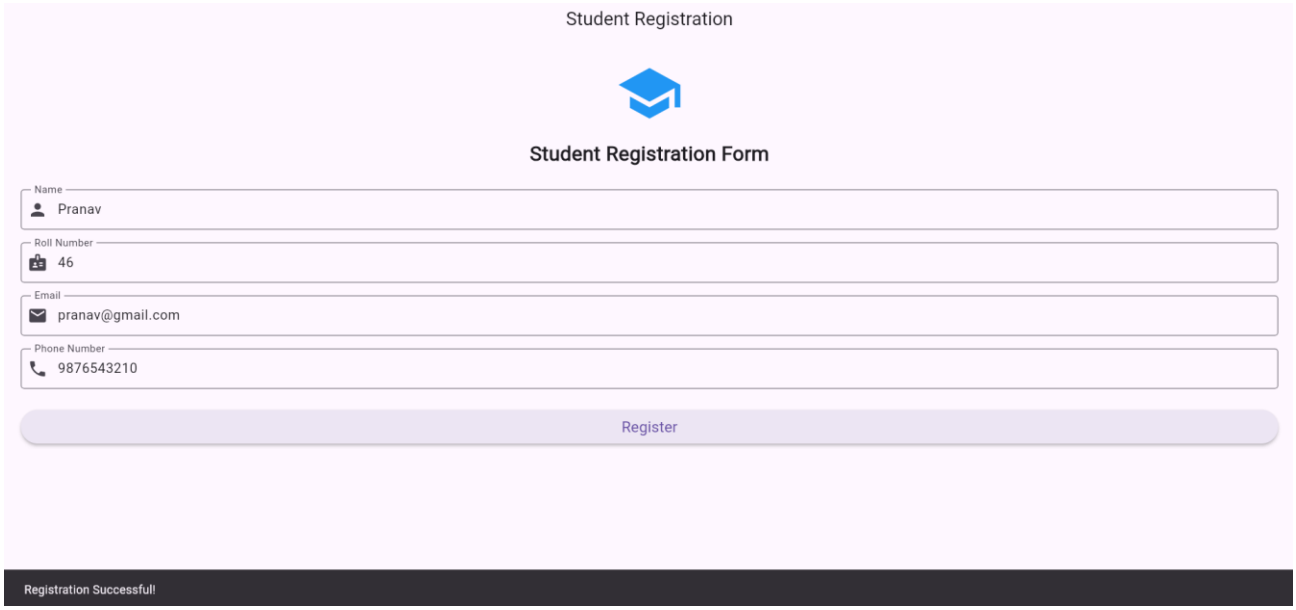
            ElevatedButton(
              onPressed: registerStudent,
              style: ElevatedButton.styleFrom(
                padding: const EdgeInsets.symmetric(
                  vertical: 15,
                ),
              ),
              child: const Text(
                'Register',
                style: TextStyle(fontSize: 18),
              ),
            ),
          ],
        ),
      ),
    ),
  );
}

@override
void dispose() {
  nameController.dispose();
  emailController.dispose();
  phoneController.dispose();
```

```
    rollNoController.dispose();

    super.dispose();
  }
}
```

## 7. Output




## 8. Result

Thus, a Flutter-based Student Registration application was successfully designed and implemented with form input fields, validation, controllers, and a registration success message.

# PROGRAM – 5

**Design the UI of an Instagram profile page using Flutter. Display the user's profile picture, username, bio, followers, following, posts count, action buttons, highlights, and a grid of images. Use appropriate Flutter widgets to build the interface.**

## 1. Aim

To design and develop a Flutter application that displays an Instagram-style profile page with profile information, statistics, highlights, action buttons, and a grid of posts.

## 2. Objective

- To design an Instagram-style user interface using Flutter.
- To display a local profile image using AssetImage.
- To display profile statistics such as posts, followers, and following.
- To create reusable widgets for profile statistics and highlights.
- To display highlights using a horizontal ListView.
- To display posts using a GridView.builder.

## 3. Software Requirements

- Flutter SDK
- Dart
- Android Studio / Visual Studio Code
- Android Emulator or Android device
- Internet connection for network images

## 4. Widgets Used

- MaterialApp – Provides the application structure.
- Scaffold – Provides the basic screen layout.
- AppBar – Displays the profile name and action icons.
- SingleChildScrollView – Makes the complete profile page scrollable.

- CircleAvatar – Displays circular profile and highlight images.
- Row – Arranges widgets horizontally.
- Column – Arranges widgets vertically.
- ListView – Displays highlights horizontally.
- GridView.builder – Displays posts in a 3-column grid.
- Image.network() – Displays images from network URLs.
- AssetImage – Displays the local profile image.
- ElevatedButton – Creates the Follow button.
- OutlinedButton – Creates the Message and add-person buttons.
- ProfileStat – Reusable widget for profile statistics.
- Highlight – Reusable widget for profile highlights.

**5. Procedure**

1. Create a new Flutter project.
2. Add the profile image as assets/profile.jpg.
3. Declare the image asset in pubspec.yaml.
4. Import the Flutter Material package.
5. Create the MyApp widget and configure the application.
6. Create the InstagramProfilePage widget.
7. Display the profile picture using CircleAvatar and AssetImage.
8. Display Posts, Followers, and Following statistics using the reusable ProfileStat widget.
9. Add username, bio, Follow, Message, and profile action buttons.
10. Create a horizontal highlights section using ListView.
11. Create reusable Highlight widgets for each highlight.
12. Add Instagram-style tab icons.
13. Display posts in a three-column grid using GridView.builder.

14.Load post and highlight images using network URLs.

15.Run the application and verify the output.

**6. Program**

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Instagram Profile',
      theme: ThemeData(
        primarySwatch: Colors.blue,
      ),
      home: const InstagramProfilePage(),
    );
  }
}

class InstagramProfilePage extends StatelessWidget {
```

```
const InstagramProfilePage({super.key});

final List<String> posts = const [
  'https://picsum.photos/300/300?1',
  'https://picsum.photos/300/300?2',
  'https://picsum.photos/300/300?3',
  'https://picsum.photos/300/300?4',
  'https://picsum.photos/300/300?5',
  'https://picsum.photos/300/300?6',
  'https://picsum.photos/300/300?7',
  'https://picsum.photos/300/300?8',
  'https://picsum.photos/300/300?9',
];

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text(
        'pranav_dev',
        style: TextStyle(
          fontWeight: FontWeight.bold,
        ),
      ),
      actions: const [
        Icon(Icons.add_box_outlined),
```

```
    SizedBox(width: 20),
    Icon(Icons.menu),
    SizedBox(width: 10),
  ],
),

body: SingleChildScrollView(
  child: Column(
    children: [

      Padding(
        padding: const EdgeInsets.all(15),
        child: Column(
          children: [

            Row(
              children: [

                const CircleAvatar(
                  radius: 45,
                  backgroundImage: AssetImage(
                    'assets/profile.jpg',
                  ),
                ),

                const SizedBox(width: 25),
```

```
    Expanded(
      child: Row(
        mainAxisAlignment:
            MainAxisAlignment.spaceAround,
        children: const [
          ProfileStat(
            number: '120',
            label: 'Posts',
          ),
          ProfileStat(
            number: '5.2K',
            label: 'Followers',
          ),
          ProfileStat(
            number: '350',
            label: 'Following',
          ),
        ],
      ),
    ),
  ],
),

const SizedBox(height: 15),
```

```
const Align(
  alignment: Alignment.centerLeft,
  child: Text(
    'Pranav R',
    style: TextStyle(
      fontSize: 16,
      fontWeight: FontWeight.bold,
    ),
  ),
),

const SizedBox(height: 5),

const Align(
  alignment: Alignment.centerLeft,
  child: Text(
    'Developer | Student\n'
    'Flutter & AI Enthusiast\n'
    'Building ideas into reality 🚀',
  ),
),

const SizedBox(height: 15),

Row(
  children: [
```

```
      Expanded(
        child: ElevatedButton(
          onPressed: () {},
          child: const Text('Follow'),
        ),
      ),

      const SizedBox(width: 8),

      Expanded(
        child: OutlinedButton(
          onPressed: () {},
          child: const Text('Message'),
        ),
      ),

      const SizedBox(width: 8),

      OutlinedButton(
        onPressed: () {},
        child: const Icon(Icons.person_add),
      ),
    ],
  ),
],
```

```
  ),
),

const SizedBox(height: 5),

SizedBox(
  height: 105,
  child: ListView(
    scrollDirection: Axis.horizontal,
    padding: const EdgeInsets.symmetric(
      horizontal: 10,
    ),
    children: const [

      Highlight(
        title: 'Projects',
        image: 'https://picsum.photos/100/100?10',
      ),

      Highlight(
        title: 'Travel',
        image: 'https://picsum.photos/100/100?11',
      ),

      Highlight(
        title: 'College',
```

```
        image: 'https://picsum.photos/100/100?12',
      ),

      Highlight(
        title: 'Friends',
        image: 'https://picsum.photos/100/100?13',
      ),

      Highlight(
        title: 'Events',
        image: 'https://picsum.photos/100/100?14',
      ),
    ],
  ),
),

const Divider(),

const Padding(
  padding: EdgeInsets.symmetric(vertical: 10),
  child: Row(
    mainAxisAlignment:
        MainAxisAlignment.spaceEvenly,
    children: [
      Icon(Icons.grid_on),
      Icon(Icons.video_library_outlined),
```

```
      Icon(Icons.person_pin_outlined),
    ],
  ),
),

GridView.builder(
  shrinkWrap: true,
  physics:
      const NeverScrollableScrollPhysics(),

  itemCount: posts.length,

  gridDelegate:
      const SliverGridDelegateWithFixedCrossAxisCount(
    crossAxisCount: 3,
    crossAxisSpacing: 2,
    mainAxisSpacing: 2,
  ),

  itemBuilder: (context, index) {
    return Image.network(
      posts[index],
      fit: BoxFit.cover,
    );
  },
),
```

```
          ],
        ),
      ),
    );
  }
}

class ProfileStat extends StatelessWidget {
  final String number;
  final String label;

  const ProfileStat({
    super.key,
    required this.number,
    required this.label,
  });

  @override
  Widget build(BuildContext context) {
    return Column(
      children: [
        Text(
          number,
          style: const TextStyle(
            fontSize: 18,
            fontWeight: FontWeight.bold,
```

```
          ),
        ),

        Text(
          label,
          style: const TextStyle(
            fontSize: 13,
          ),
        ),
      ],
    );
  }
}

class Highlight extends StatelessWidget {
  final String title;
  final String image;

  const Highlight({
    super.key,
    required this.title,
    required this.image,
  });

  @override
  Widget build(BuildContext context) {
```

```
    return Padding(
      padding: const EdgeInsets.symmetric(
        horizontal: 7,
      ),

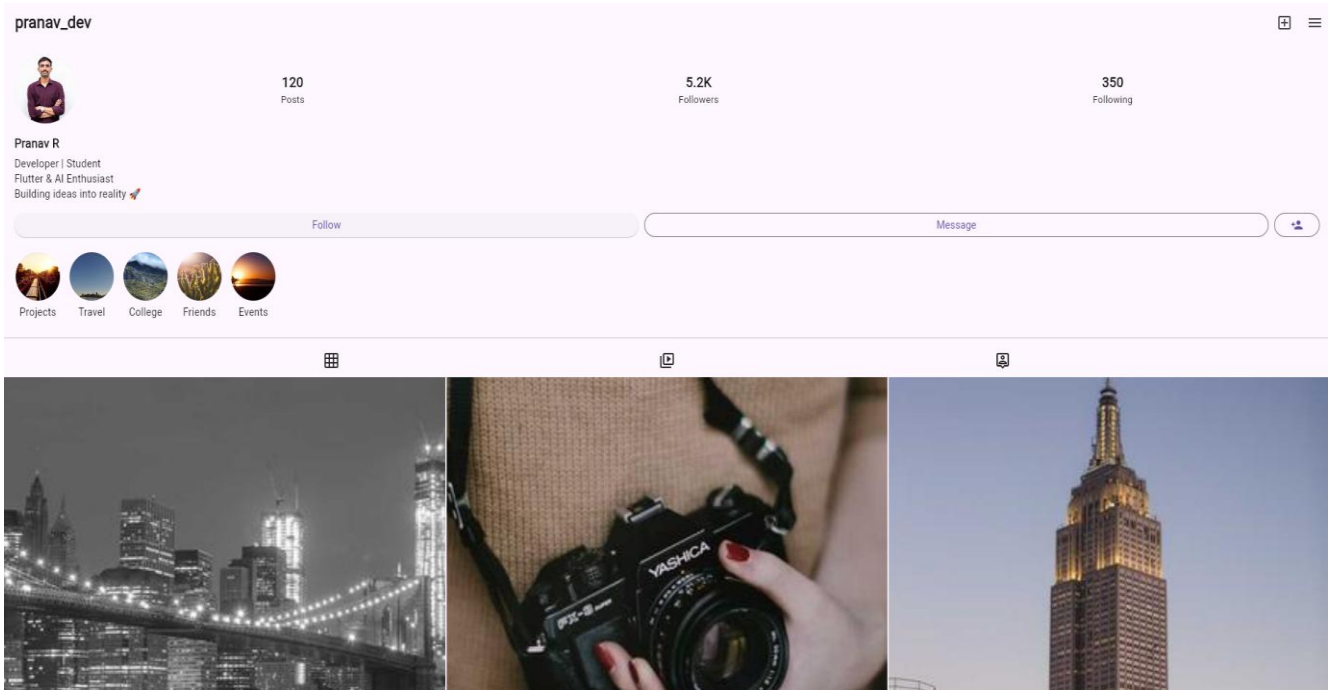
      child: Column(
        children: [

          CircleAvatar(
            radius: 32,
            backgroundImage: NetworkImage(image),
          ),

          const SizedBox(height: 5),

          Text(title),
        ],
      ),
    );
  }
}
```

## 7. Output




## 8. Result

Thus, an Instagram-style Flutter profile application was successfully designed and implemented using Flutter widgets, reusable components, local assets, network images, horizontal lists, and a three-column image grid.